Scan vs BIM — Reading List

A curated, verified reading list — to align a laser scan to a BIM model, judge every element, and put a carbon number on the difference.

Companion to the *Scan vs BIM* project · Oct 2026 · every link verified against primary sources · old foundations + current (2023–26) work

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> **How to use this.** You don't need all of it. Read §2 and the Open3D tutorial first (Phase 1), then §4–5 for the compare logic and tolerances. §6 is the IFC, BCF and viewer reference for Phases 3–4, §7–8 for the carbon layer. §9 is a lookup table, §3 is for after the baseline runs, and §10 is for framing the PhD story.

Suggested path (≈4 weeks of evenings, alongside the roadmap): Week 1 — §1 scanning basics + §2 registration, while building Phase 1 → Week 2 — §3 surveys + §5 PC2Model and Becker 2026 → Week 3 — §4 Bosché papers + USIBD LOA + §6 IfcOpenShell docs, for Phases 2–3 → Week 4 — §7–8 carbon, §6 georeferencing, then §10.

1 — Point Clouds & Laser Scanning (what the scan actually is)

Scanning Geometry: Influencing Factor on the Quality of Terrestrial Laser Scanning Points — Soudarissanane, Lindenbergh & Menenti, ISPRS J. Photogramm. Remote Sens. 66(4) (2011) paywalled
Quantifies how incidence angle and range degrade TLS points — the reason a wall scanned at a grazing angle looks "shifted" when it isn't.
`doi.org/10.1016/j.isprsjprs.2011.01.005`

Virtual Laser Scanning with HELIOS++ — Winiwarter, Esmorís Pena, Weiser et al., Remote Sensing of Environment 269 (2022) paywalled
The simulator that generated PC2Model's synthetic scans (Leica P40 profile); read it to know what noise, mixed pixels and occlusion you are being scored against.
`doi.org/10.1016/j.rse.2021.112772`

libE57: ASTM E2807 E57 Format — ASTM E57.04 / libE57 project
The vendor-neutral container PC2Model and most TLS exports use; explains what an .e57 holds (per-scan poses, intensity, colour) beyond xyz.
`libe57.org`

LAS Specification 1.4 (R15) — ASPRS (2019)
The other point-cloud format you will meet; the classification and return fields are what laspy/PDAL expose.
`asprs.org/?p=650`

Sonata: Self-Supervised Learning of Reliable Point Representations — Wu, DeTone, Frost, Shen et al., CVPR 2025
A "later" pointer: pretrained indoor point features that could separate clutter from structure before comparing scan to IFC (the roadmap's cloud-GPU item).
`arxiv.org/abs/2503.16429`

Utonia: Toward One Encoder for All Point Clouds — Zhang, Wu, Yang, Fan et al., ICML 2026
Same group's successor, one encoder across indoor, outdoor and object data; shows where segmentation priors for scan-vs-BIM are heading.
`arxiv.org/abs/2603.03283`

2 — Registration: the Method (classical baseline, in the order you will build it)

A Method for Registration of 3-D Shapes — Besl & McKay, IEEE TPAMI 14(2) (1992) paywalled
The original ICP: your fine-alignment step, and the source of its main failure mode (convergence only from a good start).
`doi.org/10.1109/34.121791`

Object Modelling by Registration of Multiple Range Images — Chen & Medioni, Image and Vision Computing 10(3) (1992) paywalled
Origin of point-to-plane ICP, which is the variant you want when the target is flat walls and slabs.
doi.org/10.1016/0262-8856(92)90066-C

Efficient Variants of the ICP Algorithm — Rusinkiewicz & Levoy, 3DIM (2001) paywalled
The taxonomy of sampling, matching, weighting and rejection choices; a checklist for the ICP settings you must justify in the write-up.
doi.org/10.1109/IM.2001.924423

Generalized-ICP — Segal, Haehnel & Thrun, RSS (2009)
Plane-to-plane ICP, a cheap upgrade over point-to-plane when the model side is sampled mesh rather than a real scan.
roboticsproceedings.org/rss05/p21.html

Fast Point Feature Histograms (FPFH) for 3D Registration — Rusu, Blodow & Beetz, ICRA (2009) paywalled
The descriptor behind your coarse step; know its radius parameters and why it struggles when scan and model densities differ.
doi.org/10.1109/ROBOT.2009.5152473

Random Sample Consensus — Fischler & Bolles, Commun. ACM 24(6) (1981) paywalled
The hypothesise-and-verify loop inside Open3D's global registration; explains the iteration count and inlier threshold you will tune.
doi.org/10.1145/358669.358692

Fast Global Registration — Zhou, Park & Koltun, ECCV (2016)
A faster, RANSAC-free alternative on the same FPFH correspondences; an easy second baseline in Open3D.
vladlen.info/publications/fast-global-registration/

Least-Squares Estimation of Transformation Parameters Between Two Point Patterns — Umeyama, IEEE TPAMI 13(4) (1991) paywalled
Closed-form similarity transform with scale: the answer to PC2Model's 0.5-1.5 random scaling on half the samples.
doi.org/10.1109/34.88573

Open3D Tutorial: Global Registration (and ICP registration) — Open3D project
The exact FPFH + RANSAC + ICP recipe from the roadmap, with voxel-size guidance; start here for Phase 1.
www.open3d.org/docs/latest/tutorial/pipelines/global_registration.html

3 — Registration: Surveys & Learned Methods (read after the baseline runs)

A Review of Point Cloud Registration Algorithms for Mobile Robotics — Pomerleau, Colas & Siegwart, Foundations and Trends in Robotics 4(1) (2015) paywalled
The best single map of ICP variants and their error models; useful for explaining why your baseline fails on specific PC2Model categories.
doi.org/10.1561/2300000035

A Comprehensive Survey on Point Cloud Registration — Huang, Mei, Zhang & Abbas, arXiv (2021)
Covers learned and optimisation-based methods including partial overlap and cross-source cases, the PC2Model situation.
arxiv.org/abs/2103.02690

TEASER: Fast and Certifiable Point Cloud Registration — Yang, Shi & Carlone, IEEE T-RO 37(2) (2021)
A robust global solver that tolerates most correspondences being outliers; a strong non-learned upgrade to RANSAC for a model-vs-scan setting.
arxiv.org/abs/2001.07715

PREDATOR: Registration of 3D Point Clouds with Low Overlap — Huang, Gojcic, Usvyatsov, Wieser et al., CVPR 2021
Learned registration designed for partial overlap, which is what a scan of part of a building against the full model is.
arxiv.org/abs/2011.13005

Geometric Transformer for Fast and Robust Point Cloud Registration (GeoTransformer) — Qin, Yu, Wang, Guo et al., CVPR 2022
The usual learned baseline to cite when you say your classical result leaves room; correspondence-free and fast.
arxiv.org/abs/2202.06688

Nothing Stands Still: A Spatiotemporal Benchmark on 3D Point Cloud Registration — Sun, Hao, Huang, Savarese et al., arXiv (2023, rev. 2025)
Registration of indoor scans of buildings under construction; the multi-epoch follow-on in the roadmap's "later" list (licence still to check).
arxiv.org/abs/2311.09346

4 — Scan-vs-BIM Foundations (as-built vs as-designed, the classic papers)

Automated Retrieval of 3D CAD Model Objects in Construction Range Images — Bosché & Haas, Automation in Construction 17(4) (2008) paywalled
The founding scan-vs-BIM paper: register scan to model, then recognise each model object by its visible surface; your per-element design in miniature.
doi.org/10.1016/j.autcon.2007.09.001

Automated Recognition of 3D CAD Model Objects in Laser Scans and Calculation of As-Built Dimensions for Dimensional Compliance Control in Construction — Bosché, Advanced Engineering Informatics 24(1) (2010) paywalled
Adds as-built dimension measurement and tolerance compliance per element, which is exactly the shifted/OK verdict.
doi.org/10.1016/j.aei.2009.08.006

Tracking the Built Status of MEP Works: Assessing the Value of a Scan-vs-BIM System — Bosché, Guillemet, Turkan & Haas, J. Computing in Civil Eng. 28(4) (2014) paywalled
Evaluates recall/precision for "is it built" and treats occlusion explicitly, supporting your not-visible vs missing split.
doi.org/10.1061/(ASCE)CP.1943-5487.0000343

Automated Progress Tracking Using 4D Schedule and 3D Sensing Technologies — Turkan, Bosché, Haas & Haas, Automation in Construction 22 (2012) paywalled
Shows how element-level verdicts roll up into progress and schedule status; the multi-epoch extension of this project.
doi.org/10.1016/j.autcon.2011.10.003

Deviation Analysis Method for the Assessment of the Quality of the As-Is BIM Models Generated from Point Cloud Data — Anil, Tang & Akinci, Automation in Construction 35 (2013) paywalled
Defines point-to-surface deviation statistics for judging a model against a scan; the metric vocabulary for your per-element CSV.
doi.org/10.1016/j.autcon.2013.06.003

Automatic Reconstruction of As-Built BIMs from Laser-Scanned Point Clouds: A Review — Tang, Huber, Akinci & Lipman, Automation in Construction 19(7) (2010) paywalled
The canonical scan-to-BIM survey; gives you the vocabulary (geometry, object recognition, relationships) for the research story.
doi.org/10.1016/j.autcon.2010.06.007

5 — Scan-vs-BIM Now & Accuracy Standard (2023-26 work, plus the tolerance spec)

PC2Model: ISPRS Benchmark on 3D Point Cloud to Model Registration — Maboudi, Harb, Ferrao et al., ISPRS Archives XLIX-B2-2026
The benchmark you score on; read for categories, transforms and the metrics the authors use (still unverified in data-sources.md).
arxiv.org/abs/2604.19596

Evaluation of ICP Variants for Point Cloud/BIM Alignment Enabling Scan-vs-BIM Comparison — Becker, Macher, Landes & Kila, ISPRS Archives XLIX-B2-2026
Same 2026 ISPRS volume, directly on ICP variants for scan-vs-BIM; the nearest published comparison to your Phase 1 ablations.
doi.org/10.5194/isprs-archives-xlix-b2-2026-1229-2026

Global BIM-Point Cloud Registration and Association for Construction Progress Monitoring — Zhang, Lu, Luo et al., Automation in Construction 168 (2024) paywalled
Registers a scan to a BIM model globally then associates points to components; a current counterpart to your align and compare stages.
doi.org/10.1016/j.autcon.2024.105796

Procedural Point Cloud Modelling in Scan-to-BIM and Scan-vs-BIM Applications: A Review — Abreu, Pinto & Matos, ISPRS Int. J. Geo-Information 12(7) (2023)
Recent review whose authors also made the CRAS dataset; good map of what scan-vs-BIM pipelines do and where they stop.
doi.org/10.3390/ijgi12070260

Labelled Indoor Point Cloud Dataset for BIM Related Applications — Abreu, Souza, Pinto & Matos, Data 8(6):101 (2023)
The data paper for the INESC CRAS scan + IFC pair: classes, scan setup and IFC provenance, to read before trusting it as ground truth.
doi.org/10.3390/data8060101

Open-Source Automatic Pipeline for Efficient Conversion of Large-Scale Point Clouds to IFC Format (Cloud2BIM) — Zbirovský & Nežerka, Automation in Construction 177 (2025)
A scan-to-IFC pipeline you can run on the same data; the reverse direction and a natural 'later' extension.
arxiv.org/abs/2503.11498

USIBD Level of Accuracy (LOA) Specification — U.S. Institute of Building Documentation (v3.1, Jan 2025)
The source for tolerance bands instead of invented thresholds; check the version PC2Model cites.
usibd.org/level-of-accuracy/

6 — BIM, IFC & Georeferencing (the model side and the viewer)

ISO 16739-1:2024 Industry Foundation Classes: Data Schema — ISO / buildingSMART (2nd ed., 2024) paywalled
The IFC standard itself; the entity definitions you will traverse when turning elements into geometry and quantities.
iso.org/standard/84123.html

IfcMapConversion (IFC 4.3.2.0) — buildingSMART International
The entity that stores eastings, northings, height and rotation; what you write to place the building on MapLibre.
standards.buildingsmart.org/IFC/RELEASE/IFC4_3/HTML/lexical/IfcMapConversion.htm

Tools for BIM-GIS Integration (IFC Georeferencing and Conversions): Results from the GeoBIM Benchmark 2019 — Noardo, Harrie, Arroyo Ohori, Biljecki et al., ISPRS Int. J. Geo-Information 9(9) (2020)
Shows which tools silently drop or misread IFC georeferencing; a warning before you trust a viewer's map placement.
doi.org/10.3390/ijgi9090502

IfcOpenShell: Geometry Processing (iterator) — IfcOpenShell docs
How to turn every IFC element into a mesh in bulk with the multithreaded iterator; the core of the compare phase.
docs.ifcopenshell.org/ifcopenshell-python/geometry_processing.html

IfcOpenShell: BCF — IfcOpenShell docs
Reading and writing BCF-XML 2.1/3.0 from Python, for the one-issue-per-deviating-element export.
docs.ifcopenshell.org/bcf.html

BCF-XML Specification — buildingSMART (GitHub)
The schema behind the issue files; check viewpoint and component-reference fields before promising 'points to the right element'.
github.com/buildingSMART/BCF-XML

That Open Engine Documentation — That Open Company
The browser IFC viewer stack for Phase 4, including how to colour elements by a property such as your verdict.
docs.thatopen.com/intro

MapLibre GL JS Documentation — MapLibre
Map rendering for the georeferenced building; you know the GIS side, this covers custom layers for placing a 3D model.
maplibre.org/maplibre-gl-js/docs/

7 — Embodied Carbon: Method (what A1-A3 means and why quantities matter)

EN 15978:2011 Sustainability of Construction Works: Assessment of Environmental Performance of Buildings — CEN (2011) paywalled
Defines the life-cycle modules; this is where your A1-A3 product-stage scope and its stated exclusions come from.
din.de/en/wdc-beuth:din21:164252701

Embodied GHG Emissions of Buildings: The Hidden Challenge for Effective Climate Change Mitigation — Röck, Saade, Balouktsi, Rasmussen et al., Applied Energy 258 (2020) paywalled
Shows embodied emissions are a large share of building impact, the motivation line for the carbon layer.
doi.org/10.1016/j.apenergy.2019.114107

Embodied Carbon Mitigation and Reduction in the Built Environment: What Does the Evidence Say? — Pomponi & Moncaster, J. Environmental Management 181 (2016) paywalled
Evidence review of what reduces embodied carbon, including reuse; frames why verified inventories help.
doi.org/10.1016/j.jenvman.2016.08.036

Critical Review of BIM-Based LCA Method to Buildings — Soust-Verdaguer, Llatas & García-Martínez, Energy and Buildings 136 (2017) paywalled
Describes how IFC quantities are mapped to LCA data and where that breaks, which is your element-to-factor step.
doi.org/10.1016/j.enbuild.2016.12.009

8 — Carbon Factors & Material Passports (pick one licence before Phase 5)

ICE Database (Inventory of Carbon and Energy) — Circular Ecology (v4.0, Dec 2024)
The widely cited UK factor set; per data-sources.md its tool use needs a paid licence, so read the terms before choosing it.
circularecology.com/embodied-carbon-footprint-database.html

EPiC Database: Usage and Licence Information — University of Melbourne (Crawford, Stephan & Prideaux)
CC BY-NC-ND with citation required; the page says what you may do, so it decides if factors can ship in a public repo.
msd.unimelb.edu.au/research/projects/current/environmental-performance-in-construction/epic-database/usage-and-licence-information

ÖKOBAUDAT — German Federal Ministry for Housing, Urban Development and Building
The official German EPD-based database (EN 15804 data), the likeliest to have an API and clear licence; check terms.
oekobaudat.de/en.html

Improving the Recycling Potential of Buildings Through Material Passports (MP): An Austrian Case Study — Honic, Kovacic & Rechberger, J. Cleaner Production 217 (2019) paywalled
BIM-based material passport in practice; the target your scan-verified inventory is a first step toward.
doi.org/10.1016/j.jclepro.2019.01.212

Materials Passports: Best Practice — Heinrich & Lang, TUM / BAMB (2019)
What information a passport holds and who needs it, which tells you which fields scan evidence can supply.
zenodo.org/record/2556515

9 — Datasets & Tools Lookup (a table you will return to)

PC2Model (v1, Nov 2025) — Zenodo, CC BY 4.0, 10.2 GB
137 samples with models, e57 scans and ground-truth transforms; 31 are houses or indoor.
zenodo.org/records/17581812

INESC TEC CRAS Lab — Zenodo, CC BY 4.0
21 TLS scans, 585M points, plus a 67 MB as-built IFC: the case study; subsample first.
`zenodo.org/records/7948116`

Open3D — Zhou, Park & Koltun (2018)
Point cloud and registration library; cite this paper when you use it.
`arxiv.org/abs/1801.09847`

CloudCompare — CloudCompare project
Visual check of your alignment and an independent cloud-to-mesh distance to cross-check your CSV.
`cloudcompare.org`

HELIOS++ — 3DGeo Heidelberg
To reproduce PC2Model-style scans or make your own occlusion test cases from an IFC mesh.
`github.com/3dgeo-heidelberg/helios`

Bonsai (BlenderBIM) Documentation — IfcOpenShell community
Free IFC and BCF viewer/editor for hand-checking elements and opening your BCF output.
`docs.bonsaibim.org`

10 — Research Framing for the PhD Story (surveys and groups)

Towards a Semantic Construction Digital Twin: Directions for Future Research — Boje, Guerriero, Kubicki & Rezgui, Automation in Construction 114 (2020) paywalled
Gives the digital-twin framing: your scan-verified model is the 'as-built state' layer such twins lack.
`doi.org/10.1016/j.autcon.2020.103179`

Construction with Digital Twin Information Systems — Sacks, Brilakis, Pikas, Xie et al., Data-Centric Engineering 1 (2020)
Open-access argument for continuous model-reality feedback; useful for the research-proposal motivation.
`doi.org/10.1017/dce.2020.16`

From BIM to Digital Twins: A Systematic Review of the Evolution of Intelligent Building Representations — Deng, Menassa & Kamat, J. Information Technology in Construction 26 (2021)
Maps the BIM to digital-twin gap and where scan data enters; a source for naming research groups to approach.
`doi.org/10.36680/j.itcon.2021.005`

* What's Still Unverified

PC2Model: the evaluation metrics the authors use and whether an evaluation server will open are not yet confirmed. The Zenodo record is labelled v1 but the file is `PC2Model_v1.4.zip`; the mismatch is unexplained.

Carbon factor licences: not decided. ICE needs a paid licence for tool use (per the earlier survey); EPiC is CC BY-NC-ND; ÖKOBAUDAT terms not read. The UNSW "ICM" database in data-sources.md could not be confirmed and is not listed here.

USIBD LOA: v3.1 (Jan 2025) is current per press releases; a direct download path was not confirmed, and PC2Model's cited version should be checked against it.

Nothing Stands Still: its dataset licence is unverified (a mirror says CC BY-NC-SA) and the paper's venue was not confirmed.

Every link verified to resolve against a primary source (Oct 2026). DOI metadata (authors, year, venue) was checked against Crossref and arXiv records; web pages were fetched or confirmed by search. Cloudflare or bot blocks prevented direct fetches of the buildingSMART, ISO, DIN, USIBD, University of Melbourne (EPiC), Zenodo and MDPI pages, so those were confirmed through search results and DOIs and are kept. libe57.org answers on plain http only (its https certificate does not match the domain). Yellow tags mark paywalled items. Dropped as unconfirmed: the UNSW ICM database. Left out for lack of a verifiable primary source: a NIST-linked scan-to-BIM survey. Companion document: *3D Reality Capture & BIM — Foundations*.